

Measurement System Analysis (MSA)

Duration: 1 Day

Mode: Online / Classroom / On-Site

Certification: QLSS Certificate of Completion



Course Introduction

Measurement System Analysis (MSA) is a statistical technique used to assess the accuracy, repeatability, and reproducibility of measurement systems in manufacturing and quality environments. It is a foundational core tool in the IATF 16949 quality management system and essential for effective decision-making based on measured data.

This training introduces the key concepts of bias, linearity, stability, and Gage R&R, providing hands-on understanding of how to analyze and improve measurement systems. The session emphasizes interpretation, action thresholds, and real-life application using practical case studies.



Pre-requisites

- Basic understanding of quality control or statistical concepts
- Familiarity with production or inspection environments
- No prior MSA training required



Learning Objectives

- Understand the importance of MSA within the APQP and SPC framework
- Identify sources of variation in measurement systems
- Perform Gage Repeatability & Reproducibility (R&R) studies
- Evaluate bias, linearity, and stability of instruments
- Interpret MSA results and determine system adequacy
- Apply acceptance criteria based on AIAG MSA Manual (4th/5th Edition)
- Improve decision-making by validating measurement processes



Course Agenda

1. Introduction to MSA

- Purpose of measurement system validation
- Role of MSA in IATF 16949 and core tools
- When MSA is required (e.g., new gages, process changes)

2. Key Measurement System Concepts

- Precision vs. accuracy
- Types of measurement errors
- Variation: part-to-part, operator, equipment, environment

3. Types of MSA Studies

- Variable MSA (continuous data)
- Attribute MSA (pass/fail, go/no-go)
- Special study types: Bias, Linearity, Stability

4. Gage R&R Study – Variable Data

- Components: Repeatability and Reproducibility
- Short method (ANOVA and Xbar-R methods)
- Acceptability criteria (% R&R, ndc, % contribution)

5. Attribute Agreement Analysis (AAA)

- Operator agreement, correctness vs. consistency
- False positives and false negatives
- Application in inspection and audits

6. Bias, Linearity, and Stability

- How to detect and interpret instrument bias
- Use of reference standards
- Monitoring linearity across range of measurement
- Long-term stability of gages

7. Using MSA Results for Action

- Making go/no-go decisions based on metrics
- When to recalibrate or replace instruments
- Linking MSA to Control Plan and audit trail



What You'll Receive

- QLSS Certificate of Completion
- Training Material / Handout (PDF Format)
- Final Assessment with Individual Evaluation Report



Who Should Attend

- Quality Engineers and Technicians
- Metrology and Calibration Personnel
- Process Owners and Inspection Supervisors
- Internal Auditors and Core Tool Champions
- APQP and SPC Implementation Teams



Trainer Profile

Our trainers bring deep industry knowledge in statistical quality control, metrology, and audit practices. They have led MSA implementations across diverse industries and are skilled in translating complex statistical outputs into actionable process decisions. Sessions are conducted with real-life examples and expert facilitation.



Instructional Methodology

- Real-world case studies and calculator-based examples
- Step-by-step MSA worksheet guidance
- Interpretation of AIAG acceptance limits
- Group exercises with measurement data
- Visual walkthrough of ANOVA and Xbar-R output



What Makes QLSS MSA Training Unique

- Simplified explanation of complex statistical methods
- Focus on interpretation and decision-making
- Side-by-side analysis of variable vs. attribute MSA
- Ready-to-use templates and visual guides
- Covers both AIAG 4th and 5th edition highlights

